

<https://www.adafruit.com/product/2652?srltid=AfmBOooP2HxngSyt8EumHzuZo6vPtVJa m8zxM0ovIFUBWzYmqusAkces>

Adafruit BME280 I2C or SPI Temperature Humidity Pressure Sensor - STEMMA QT

Bosch has stepped up their game with their new BME280 sensor, an environmental sensor with temperature, barometric pressure and humidity! This sensor is great for all sorts of indoor environmental sensing and can even be used in both I2C and SPI!

This precision sensor from Bosch is the best low-cost sensing solution for measuring humidity with $\pm 3\%$ accuracy, barometric pressure with ± 1 hPa absolute accuracy, and temperature with $\pm 1.0^{\circ}\text{C}$ accuracy. Because pressure changes with altitude, and the pressure measurements are so good, you can also use it as an altimeter with ± 1 meter or better accuracy!

The BME280 is the next-generation of sensors from Bosch, and is the upgrade to the BMP085/BMP180/BMP183 – with a low altitude noise of 0.25m and the same fast conversion time. It has the same specifications, but can use either I2C or SPI. For simple easy wiring, go with I2C. If you want to connect a bunch of sensors without worrying about I2C address collisions, go with SPI.

Nice sensor right? So we made it easy for you to get right into your next project. The surface-mount sensor is soldered onto a custom made PCB and comes with a 3.3V regulator and level shifting so you can use it with a 3V or 5V logic microcontroller without worry. The breakout is made in the [STEMMA QT form factor](#), making them easy to interface with. The [STEMMA QT connectors](#) on either side are compatible with the [SparkFun Qwiic](#) I2C connectors. This allows you to make solderless connections between your development board and the BME280 or to chain it with a wide range of other sensors and accessories using a [compatible cable](#).

Planned board to use on the Athena 2.0 Pro max: arduino 5

<https://pypi.org/project/adafruit-circuitpython-bme280/>

<https://docs.ros.org/en/jazzy/Tutorials/Beginner-Client-Libraries/Writing-A-Simple-Py-Publisher-And-Subscriber.html>

<https://ubuntu.com/tutorials/gpio-on-raspberry-pi#1-overview>